Testing and Evaluation

(ENG520)

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Lesson 01

CONCEPT OF ASSESSMENT-I

Topic- 001: Measurement, Assessment and Evaluation

Measurement

Measurement is the process by which the attributes or dimensions of some object (both physical and abstract) are quantified. The tools used for this purpose may include test, observation, checklist, homework, portfolios, project etc. Measurement can be easily understood if we use this word to measure height and distance because these things are physical present in the existence so height can easily be measured by scale. But in the field of education, our variable are not physical and cannot be directly measured e.g. attitude, behavior, and achievement etc. these all are abstract, so there measurement is relatively difficult than those who have physical existence. The tool used for measuring the abstract variables cannot measure exactly like scale (thermometer). So, in this whole course, whenever the measurement word is used it means that tool which will be used for measuring student abilities and then converts it in numerical form.

Assessment

It means an appraisal of something to improve the quality of teaching and learning process for deciding what more can be done to improve the teaching, learning and outcomes.

Evaluation

Evaluation is the process of making a value judgment against intended learning outcomes and behavior, to decide the quality and extent of learning. Evaluation is always related to your purpose, you aligned your purpose of teaching with what students achieved at the end, with their quality and quantity of learning.

Example:

In a classroom situation, as a teacher, when you teach a chapter or unit to a class, first you made the objectives, either you will do it yourself as a teacher or you take it from a curriculum document of the particular subject. Objectives are also written at the start of the chapter or book which shows that at the end of the unit what the student will be able to do this, that is also referred as student learning outcomes. These SLOs can be checked by two aspects:

1. Assessment 2. Evaluation.

When you, as a teacher reflect on your teaching on a daily basis that what you teach yesterday was a good way to teach or not, what i taught was the students need, did they understand what I teach, and then you can decide changing which things can cause the improvement in the learning process of students. This is known as ASSESSMENT. At the end of the assessment, opinion is not about the individual, it is about the process from which the students or individuals are passing for the betterment of that process. Evaluation is that when the learning process is complete and you want to see what are your targets or objectives and how much my students achieved those objectives, then the tools are made and measures come from that tell that how much your student learn and what is the quality of their learning.

Difference in Measurement, Assessment and Evaluation

These terms are not different words for same concept, but a process, serving as prerequisite to each other and having unique purpose. Every mechanism or process starts with the measurement. In the field of education, measurements generate tools, i.e. test, observation, quiz, checklist, homework and portfolios. They can give information about students' learning. The information we got from the tool is MEASUREMENT. Now if that measurement is used for the process of making the teaching learning process better than it is ASSESSMENT.

EVALUATION

It is a systematic determination of a subject's merit, worth and significance, using criteria governed by a set of standards

Assessment and Evaluation don't exist in a hierarchy; they both are parallel and different in purpose. Measurement is the source to move towards assessment and evaluation because it provides the base and evidence to quantify the teaching learning process. The quantified number has no meaning until we do an assessment or evaluation. Assessment purpose is to make the teaching learning process better so that student learning improves and measurement purpose is to align the learning with a purpose.

Topic – 002: Classroom Assessment

The process of gathering, recording, interpreting, using and communicating information about a child's progress and achievement during the development of knowledge, concepts, skills and attitudes is called classroom assessment.

When we are teaching in a classroom, we are doing four things:

1. Developing students 'knowledge
2. Improving their concepts
3. Teaching the skills
4. Making their attitudes

And while doing so, we collect information about the development of these things, and then we record, interpret and use it to communicate about the learning progress of students. This whole procedure is called classroom assessment **Classification of Assessment**

Assessment can be classified in four ways

1. Nature of Assessment
2. Format of Assessment
3. Use in Classroom Instruction
4. Method of Interpreting Results

1. Nature of Assessment

i. Maximum Performance Assessment
ii. Typical Performance Assessment

2. Format of Assessment

i. Fixed Choice Assessment ii. Complex Performance Assessment **3. Use in Classroom Instruction**

i. Placement Assessment ii. Formative Assessment iii.

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Diagnostic Assessment iv. Summative Assessment

4. Method of Interpreting Results

i. Norm Referenced Assessment
ii. Criterion Referenced Assessment

Topic – 003: Types of Assessment

1. Types of assessment by nature
2. Types of assessment by format

1. Types of Assessment by Nature

i. Maximum Performance Assessment ii. Typical Performance Assessment

Maximum Performance Assessment

Maximum performance assessment determines what individuals can do when performing at their best e.g. assess student in an environment when they exhibit their best performance. Procedure of this type is concerned with how well individual perform when they are motivated to obtain as high score as possible. This type of assessment includes aptitude tests and achievement tests. In achievement test, the student learn by themselves or we teach them and at the end, we want to see that how much student learn against our target so in this situation we make a test from which we can determine their best abilities. It is designed to indicate the degree of success in any past learning activity.

We use aptitude test when we want to predict the success in future learning activity, e.g. it is used when we want to see the interest of students in a particular field like medicine, sport and teaching. We know that different abilities used for going in different professions, we make a test depending on these abilities and then try to assess, in what abilities the students perform well.

Typical Performance Assessment

The second category is a typical performance assessment determines what an individual will do under natural conditions. This type of assessment includes attitude, interest, personality inventories, observational techniques and peer appraisal. Here the emphasis is on what students will do rather than what they can do.

2. Types of Assessment by Format

i. Fixed Choice Assessment
ii. Complex Performance Assessment

Fixed Choice Assessment

Fixed choice assessment is sued to measure the skills of people efficiently (means measure more skills in less time) and for this we usually use fixed choice items, i.e. multiple choice question, matching exercises, fill in the blanks and true false. It is called fixed choice because the person who is attempting the paper does not need to write the answer, just need to choice the answer. From these, we can assess

student abilities of lower level learning. Fixed choice assessment is used for efficient measurement of knowledge and skills.

Complex Performance Assessment

Complex performance assessment is used for measurement of performance in contexts and the problems valued in their own right. This includes hands on laboratory experiments, projects, essays, oral presentations. E.g. if want to measure the student's ability of writing an essay and this cannot be judged by fixed response items

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Lesson 02

CONCEPT OF ASSESSMENT-II

Topic- 001: Use of Assessment in Classroom Instruction

Classification of assessment in terms of its uses in classroom instruction

i. Placement Assessment
ii. Diagnostic Assessment

Placement Assessment

Placement assessment determines prerequisite skills, degree of mastery of course goals and mode of learning. It is used when we want to assess student's prior knowledge so that we can decide what the level of student is. It is associated with a student's entry level performance to know either student have a sufficient knowledge required for a particular course or not. Through placement assessment, teacher can be able to know that where a student should be placed according to the present knowledge or skills. It determines the level of student knowledge at the beginning of session and helps teacher plan the lesson accordingly. In the classroom, the teacher can use placement assessment to assess the level of students' knowledge and skills and then make lesson plans keeping in mind the level and need of students accordingly. It also determines the interest and aptitude of the student regarding a subject and helps in selecting the correct path for the future.

Examples:

Readiness test: It is used to determine the students' knowledge or concept about a particular course of instruction or what is the level of students.
Aptitude test: It is used for the admission in a particular program.
Pretest: It is made according to the course objectives and determines the student present knowledge about them.
Self- report inventories: It determines the student level by interviewing or discussion.

Diagnostic Assessment

It determines the causes (intellectual, physical, emotional, environmental) of persistent learning difficulties. E.g. if you are having a headache, first you will try to cure it by yourself by taking a medicine and you got a relief, but if you did not get a relief by taking the medicine then either you change your medicine or you go to the physician or doctor. At first, doctor prescribed medicines, if you still have a headache you again go to the doctor, then the doctor suggests you the tests, i.e. blood test, urine test etc. Then finally by seeing the test reports the doctor able to recognize the reason or cause of the headache. And when doctor knows the root of your headache then he will prescribe you the medicine for that cause, this is diagnosis.

A diagnosis does not start at the first day, it is for the constant or continuous problems, e.g. if a student continues to experience failure in reading or mathematics or any other subject despite the use of prescribed alternative methods , then a diagnosis is indicated. Teachers try to find out what is the root of students' failure.

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Topic- 002: Use of Assessment in Classroom Instruction – Formative and Summative Assessments

Formative Assessment:

It determines learning progress, provides feedback to reinforce learning, and correct learning errors. When we assess student during classroom instruction with a purpose to have a feedback that how can we make our teacher learning process better, that is formative assessment. In this assessment, we are not assessing what students learnt or not, rather we assess the process behind the students learning .The process behind the student learning includes a teaching method, book, if we make all these things according to the needs of students then learning will improve.

It is conducted during the academic session or teaching-learning process so that I can get a feedback about my way of teaching and how students are learning and decisions are made on the basis of results immediately. It is an ongoing process to modify teaching strategies on the basis of students need.

It provides feedback to teachers:

- About weakness and strength of learning process
- To modify their teaching practices
- To improve teacher-learning process

It also helps students to reflect on their weaknesses and encourages them for their successful learning. Formative assessment provides feedback to students who are struggling with a specific content area or concept. The main difference between formative and summative assessment is that the in formative assessment, improvement is in the process of learning rather than to certify students .We use different tools for formative assessment and it includes teacher made tests, custom made tests from textbook.

Summative Assessment

It comes at the end of the instructional session (course of the unit). It is designed to measure extend of achievement of intended learning outcomes. The primary utility of this type of assessment is to assign grades and certifying the level of mastery and expertise in a certain subject. It is usually done through teacher made achievement tests or alternative assessment techniques like portfolio to summarize the overall performance of the student at the end of the session. It is not compulsory to do it at the end of the semester, in semester system, there is a midterm and final term in one semester, and these both are summative assessment. It usually compares the student learning either with other students' learning (norm-referenced) or the standard for a grade level (criterion-referenced). Summative assessment includes teacher made survey test, performance, rating scales and product scales.

Topic- 003: Types of Assessment: Methods of Interpreting Results

- Norm-referenced Assessment
- Criterion-referenced Assessment

Norm-referenced Assessment

Norm-referenced assessment measures the students' performance, according to the relative position in any known group. E.g. ranks tenth in classroom group of 50 or top 5 students in a class. Relative position means a point defined with reference to another position (where a student stands compared to other students) rather than reporting students' achievement, it reports students standing among other students. NRT is utilized to discriminate between a certain groups of students. It is never used for certification or

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issuing grades to students. The position of the student is generally represented by percentile scarce noting the percentage of students achieving same or low scores in the test.
Examples of NRT are NTS or CSS exams. E.g. if you achieved a 97% percentile on NTS test, it means there are 96% who scored lower than you.

Norm referenced test includes items with average difficulty and high discriminating power. This provides a large spread of scores which makes it easy to declare the relative positions of students. Because the purpose of NRT is not to certify grades, so the test made form it must be of average difficulty means test items should not be very easy or very tough. If all the items are difficult then no student will be able to solve it and we are not able to discriminate who are good students. And if all the items are easy then even the lower ability students can solve it then we are not able to discriminate it. Norm referenced test includes standardized aptitude, achievement tests, teacher-made survey tests, interest inventories, adjustment inventories.

Criterion-referenced Assessment

Criterion-referenced assessment describes student performance, according to a specific domain of clearly defined learning tasks, e.g. adds single-digit whole numbers. With this you do not compare student's performance with other students rather you compare the performance of all students with criteria (in our case that criteria are our learning outcomes). It is most commonly used in schools to report the achievement of learning outcomes against set goals rather than other students. It grades the students to pre-defined criteria and student's grades represent their mastery over content. Students with the same level of expertise achieve the same level of grades. A cut point is determined to distinguish between failed and successful students, regardless of score of highest and lowest achiever. It consists of teacher-made tests, custom made tests from the test publishers and observational techniques.

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Lesson 03

ASSESSMENT, TESTING AND NATIONAL CURRICULUM

Topic- 007: Role of National Curriculum in Assessment

In national curriculums of Pakistan, learning of student is classified into four levels.

1. Competency
2. Standards
3. Benchmarks
4. Student learning outcome (SLOs)

Competency

It is a key learning area, for example, algebra, arithmetic, geometry, etc. in mathematics and vocabulary, grammar, composition, etc. in English.

Standards

These define the competency by specifying broadly, the knowledge, skills and attitudes that students will acquire, should know and be able to do in a particular key learning area during twelve years of schooling.

Benchmarks

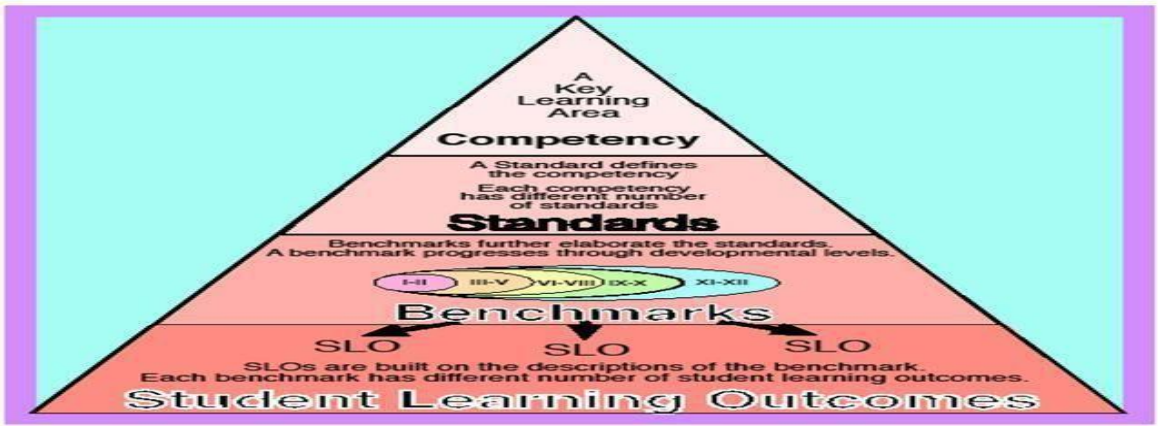
The benchmarks further elaborate the standards, indicating what the students will accomplish at the end of each of the five developmental levels in order to meet the standard.

Student Learning Outcomes

These are built on the descriptions of the benchmarks and describe what students will accomplish at the end of each grade. It is the lowest level of hierarchy.

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Topic- 008: Connecting all Four Levels in Curriculum



In the image above, SLOs are at the bottom which is the lowest level. All SLOs combined to make a benchmark and benchmarks convert into standards and then into competency.

Example:

Example is taken from curriculum of English subject.

Competency 1: Reading and thinking skills

Standard 1: All students will discover and understand a variety of text types through tasks which require multiple reading and thinking strategies for comprehension, fluency and enjoyment.
Benchmark 1: Use reading readiness strategies **Student learning outcome:**

1. Articulate, identify and differentiate between the sounds of individual letters, digraphs and trigraphs in initial and final positions in a word.
2. Identify paragraph as a graphical unit of expansion, ¾ know that word in a sentence join to make sense in relation to each other.

Topic- 009: Modes of Assessment in Curriculum

Curriculum document provides specific guidelines for assessment.
Example:
6.1. The two forms of assessment recommended are:

6.1.1. Periodic/formative assessment through homework, quizzes, class tests and group discussions.
6.1.2. End of term/ summative assessment through final examination.

Purpose of Assessment and Curriculum-English 2006

The assessment system for the present curriculum should include:

- A clear statement of the specific purpose(s) for which the assessment is being carried out.
- A wide variety of assessment tools and techniques to measure student's ability to use language effectively.
- Criteria to be used for determining performance levels for the SLOs for each grade level.
- Procedures for interpretation and use of assessment results to evaluate the learning outcomes.

Form of Suitable Assessment Tools- English 2006

- MCQs
- Constructed response
  - Restricted response
  - Extended response
- Performance tasks

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Lesson 04

TAXONOMIES OF EDUCATIONAL OBJECTIVES AND ASSESSMENT –I

Topic- 010: Taxonomies of Educational Objectives and Assessment

Every assessment regardless of its purposes rests on three important pillars:

1. A model for how students present knowledge and develop competence in the subject domain
2. Tasks or situations that allow the examiner to observe the students' performance
3. Inferences from performance evidence about the quality of learning

In developing a test to assess student learning, taxonomy provides a framework of categories with different hierarchical levels of outcomes.

Popular Taxonomies

1. Bloom's taxonomy of educational objective
2. Structure of Observed Learning Outcomes (SOLO)
3. Depth of Knowledge (DOK)

Topic- 011: Bloom's Taxonomy and SOLO Taxonomy

The taxonomy of structure of observed learning outcomes (SOLO) was initially developed by Biggs and Collis in 1982, and then well described in Biggs and Tang in 2007. It carries five different levels of competency of learners.

Levels of Structure of Observed Learning Outcomes (SOLO)

1. Pre-structural
2. Uni-structural
3. Multi-structural
4. Relational
5. Extended Abstract

Depth of Knowledge (DOK)

DOK (Depth of Knowledge) was presented by Webb in 1997, giving four levels of learning activities.

Levels of Depth of Knowledge (DOK)

1. Recall
2. Skill/Concept
3. Strategic Thinking
4. Extended Thinking

Bloom's Taxonomy of Learning Objectives Bloom's Taxonomy was presented by Benjamin Bloom in the 1956, consists of a framework with most common objectives of classroom instructions.

1. Cognitive
2. Affective

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3. Psychomotor

Cognitive Domain

- Knowledge
- Comprehension
- Application
- Analysis
- Synthesis
- Evaluation

Affective Domain

- Receiving
- Responding
- Valuing
- Organization
- Characterization

Psychomotor Domain

- Perception
- Set
- Guided Response
- Mechanism
- Complex covert Response
- Adaption
- Origination

Topic- 012: SOLO Taxonomy

Levels of SOLO

1. Pre-structural
2. Uni-Structural
3. Multi-structural
4. Relational
5. Extended Abstract

1. Pre-structural

Students are simply able to acquire bits of unconnected information and respond to a question in a meaningless way.

Example:

Question: What is your name?

Answer: What is your name?

2. Uni-Structural

Student shows concrete understanding of the topic. But at this level is only able to respond one relevant element from the stimuli or an item that is provided. Indicative verbs: identify, memorize, do simple procedure

3. Multi- Structural

The student can understand several components, but the understanding of each remains discreet. A number of connections are made, but the significance of the whole is not determined. Ideas and concepts around an issue are disorganized and aren't related together.

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Indicative verbs: enumerate, classify, describe, list, combine, do algorithms.

4. Relational

At this level, the learner is able to understand the significance of the parts in relation to the whole. Ideas and concepts are linked, and they provide a coherent understanding of the whole.

Indicative verbs: compare/contrast, explain causes, integrate, analyze, relate, and apply.

5. Extended Abstract

At this level, the learner is able to think hypothetically and can synthesize a material logically. Student make connections not only within the given subject area, but also understanding is transferable and generalizable to different areas.

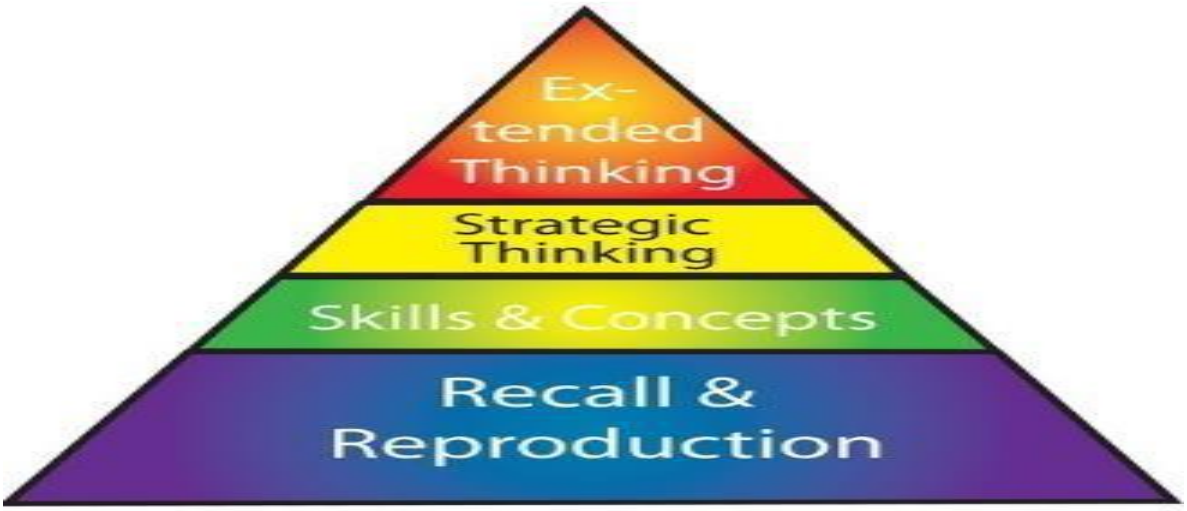
Indicative verbs: theorize, generalize, hypothesize, reflect, generate.

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Topic- 013: Depth of Knowledge

Levels of DOK

1. Recall
2. Skill/concept
3. Strategic Thinking
4. Extended Thinking



DOK measures the degree to which the knowledge brings about from students on assessments is as complex as what students are expected to know and do as stated in the curriculum.

Levels of DOK (Depth of Knowledge) Recall

Recall of a fact, information, or procedure. The subject matter at this particular level usually involves working with facts, terms and/or properties of objects.

Key words: list, enlist, name, define etc.

Skill/Concept

It includes the engagement of some mental processing beyond recalling or reproducing a response. Use

information or conceptual knowledge, two or more steps, not just recalling Key words: Graph, separate, relate, contrast, narrate, compare etc.

Strategic Thinking

Items falling in this category demand a short-term use of higher order thinking processes, such as analysis and evaluation, to solve real-world problems with predictable outcomes.

Key words: Argue, critique, formulate

Extended Thinking

Learning outcomes to this level demand extended use of higher order thinking processes such as synthesis, reflection, assessment and adjustment of plans over time. Key words: create synthesize, design and reflection.

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Lesson 05

TAXONOMIES OF EDUCATIONAL OBJECTIVES AND ASSESSMENT –II

Topic- 014: Bloom's Taxonomy-I

There are three main domains of learning and all teachers should know and use them to construct lessons.

1. Cognitive Domain
2. Affective Domain
3. Psychomotor Domain

1. Cognitive or Thinking Domain

In 2000-01 revisions to the cognitive taxonomy were spearheaded by one of Bloom's former students, Lorin Anderson, and Bloom's original partner in defining and publishing the cognitive domain, David Krathwohl. One of the major changes that occurred between the old and the newer updated version is that the two highest forms of cognition have been reversed.

Levels of Cognitive Domain Old Cognitive Domain Knowledge

It is defined as the remembering of previously learned material. This may involve the recall of a wide range of facts, procedures, principals and generals, the recall of the procedures and the processes. Sample Question: Define the 6 levels of Bloom's taxonomy of the cognitive domain.

Comprehension

It is defined as the ability to grasp the meaning of the material. An individual can make use of the content or idea being communicated without necessarily related it to other content and seeing its fullest implications. Sample Question: explain the purpose of Bloom's taxonomy of the cognitive domain.

Application

It refers to the ability to use the previously learned material in new and concrete situations. The abstractions may be in the shape of universal ideas, rules of methods. Sample Question: write an instructional objective for each level of Bloom's taxonomy.

Topic- 015: Bloom's Taxonomy-II

Analysis

The breakdown of a concept into its constituent parts such that the relative hierarchy of the concept is made easy to understand or the relation between the parts of the concept is elaborated. Sample Question: compare and contrast the cognitive and affective domains.

Synthesis

There is a collection of the constituents or parts of a concept so as to make a whole. This is a stage when an individual is working with the pieces and grouping them in such a way as to formulate a pattern or structure not clearly there before. Sample Question: Design a classification scheme for writing educational objectives that combines the cognitive, affective, and psychomotor domains.

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Evaluation

It is concerned with the ability to judge the value of the material for a given purpose. Judgments are made on the definite criteria. Sample Question: How far the different BISEs and universities are developing papers using Bloom's taxonomy? Support your answer with arguments.

Topic- 016: Revised version of Bloom‘s Taxonomy

The revised cognitive domains levels are:

Remembering

Exhibit memory of previously learned material by recalling facts, terms, basic concepts, and answer

Key verbs

Choose, define, find, how, label, list, match, name, omit, recall, relate, select, show, spell, tell, what, when, where, which, who, why

Understanding

Constructing meaning from different types of functions be they written or graphic messages, or activities.
Key verbs: Classify, compare, contrast, demonstrate, explain, extend, illustrate, infer, interpret, outline, relate, rephrase, show, summarize, and translate

Applying

Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.
Key verbs:, Apply, build, choose, construct, develop, experiment with, identify, interview, make use of, model, organize, plan, select, solve and utilize.

Analyzing

Breaking materials or concepts into parts, determining how the parts relate to one another, or how the parts relate to an overall structure or purpose.
Key verbs: Analyze, assume, categorize, classify, compare, conclusion, contrast, discover, dissect, distinguish, divide, examine, function, inference and inspect

Evaluating

Making judgments based on criteria and standards through checking and critiquing.
Key verbs: Agree, appraise, assess, award, choose, compare, conclude, criteria, criticize, decide, deduct, defend, determine, disprove and estimate

Creating

Putting elements together to form a coherent or functional whole; reorganizing elements into a new pattern or structure through generating, planning, or producing.
Key verbs: Adapt, build, change, choose, combine, compile, compose, construct, create, delete, design, develop, discuss, elaborate, estimate and formulate

These categories range from simple to complex and from concrete to the abstract level of student‘s learning. It is assumed that the taxonomy represents a cumulative hierarchy, so that mastery of each simpler category is considered as a prerequisite to mastery of the next, more complex one.

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Comparison of Bloom, SOLO and DOK

A comparison of hierarchical division in three taxonomies of learning objectives

SOLO Taxonomy	Bloom's Taxonomy	DOK Taxonomy
Level 1: Pre-structural Level 2: Uni-structural	Level 1 Knowledge	Level 1:REcall
Level 3:Multi-structural	Level 2:Comprehension	Level 2: Skill/Concept
Level 4: Relational	Level 3: Application	Level 3 Strategic Thinking
Level 5: Extended Response	Level 4: Analysis Level 5: Synthesis Level 6: Evaluation	Level 4: Extended Thinking

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Topic- 017: Instructional Objectives

Instructional Objectives as Learning Outcome

Instructional goals and objectives are stated in terms of actions to be taken. When viewing instructional objectives in terms of learning outcomes, we are concerned with products rather than process of learning.

Sources for Lists of Objectives

- Professional Associations standards
- State Content Standards
- Methods Books
- Year Books
- Encyclopedia of Educational Research
- Curriculum Frameworks
- Test Manuals

Criteria of Selecting the Final List of Objectives

Prepare tentative list of instructionally relevant learning outcomes Review the list for:

- Completeness
- Appropriateness
- Soundness
- Feasibility

General Objectives

- Stating the general objectives is selecting the proper level of generality.
- Objective should be specific enough to provide the direction for instruction but not so specific that instruction is reduced to training

- Stating general objectives in general terms, we provide for the integration of specific facts and skills into complex response
- General statements gives teachers freedom in selecting the method and materials of instruction List of general objectives shows the desired level of generality
- Knows basic terminology
- Understand concepts
- Relates concepts to everyday observations
- Applies principles to new situations
- Interpret graphs
- Demonstrate scientific attitude

Specific Learning Outcomes

Each General objective must be defined by a sample of specific learning outcome to clarify how students can demonstrate that they have achieved general objective. Until the general objective are further defined in this manner they will not provide adequate direction for assessment

Steps for Stating Specific Outcomes

- List below each general objective a representative sample of specific learning outcome that describes terminal performance students are expected to demonstrate.
- Begin each specific learning outcome with an action verb that specifies observable performance.
- Make sure that each specific learning outcome is relevant to the general objective it describes.
- Include enough SLOs to describe adequately the performances of students who have attained the objectives.
- Keep the SLOs sufficiently free of course content so that the list can be used with various units of study.
- Consult reference materials for the specific components of those complex outcomes that are difficult to define.
- Add a third level of specificity to the list of outcomes, if needed.

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Lesson 06

PURPOSE OF TESTING-I

Topic- 018: Educational Decisions Making

Types of Educational Decisions

- Instructional
- Grading
- Diagnostic
- Selecting
- Placement
- Counseling and Guidance
- Program or Curriculum
- Administrative

These types of decisions are taken at different levels. Some are decided at the board/ administrative level while some are taken at school management level and other are taken in classrooms by teachers.

Instructional Decisions

Instructional decisions are the nuts and bolts types of decisions made in the classroom by teachersSuch decisions include deciding to:

- spend more time on specific units
- regroup student in class for better management
- instructional plans

Grading Decisions

Educational decisions based on grades are also made by the classroom teacher but much less frequently than instructional decisions. For the most students grading decisions are the most influential decision made about them.

Diagnostic Decisions

Diagnostic decisions are those made about a student‘s strengths and weaknesses and the reasons behind them. Teachers make diagnostic decisions based on information yielded by an informal teacher made test decisions about diagnostic nature can also be made with the help of standardized tests.

Selection Decisions

Selection decisions involve test data used in part for accepting or rejecting applicants for admission into a group, program, or institution.

Placement Decisions

Placement decisions are made after an individual has been accepted in a program. They determine where in a program someone is the best suited to begin with.

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Program or curriculum decision

This type of decision is taken at policy level. Where it is decided if a lesion, unit or subject will continue or abandoned for the next academic session according to the national objectives of education.

Administrative Decisions

Administrative policy decisions may be made at school, district, state or national level. Based on measurement data, it includes financial decisions of schools.

Topic- 019: Types of Test

In classroom assessment different forms of assessments are utilized. Each form of test has its own benefits and disadvantages. The most common type of assessment used in classrooms is written assessment.

Types of Written Tests

- Verbal
- Non-verbal
- Objective
- Subjective
- Teacher Made
- Standardized
- Power
- Speed

Verbal

Emphasize reading, writing, or speaking. The most tests in education are verbal tests.

Non-verbal

It does not require reading, writing or speaking ability. The tests composed of numerals or drawings are an example of non-verbal test.

Objective

It refers to scoring of tests when two or more scorers can easily agree on whether the answer is correct or incorrect; the test is an objective one. True/ false, multiple choice and matching tests are examples of it

Subjective

When it is difficult for two scorers to agree on whether an item is correct or incorrect, the test is a subjective one. Essay tests are the example of it.

Teacher Made

It is constructed solely by teacher only to be used in the classroom. This type of test is custom designed according to need and issues related to specific class.

Standardized

Test constructed by measurement experts over a period of years. They are designed to measure broad national objectives and have a uniform set of instructions that are adhered to during each administration. Mostly it has tables of norms, to which a student performance may be compared to determine where the student stands in relation to a national sample of students at the same level of age or grade

Power

Tests with liberal time limits that allow each student to attempt each item. Items tend to be difficult

Speed

Tests with time limits so strict that no one is expected to complete all items. Items tend to be easy.

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Topic- 020: Norm Referenced Assessment (NRT)

General purpose of assessment is to gather information to make better and more informed decision. The utility of that information is what differentiate among thetypes of assessments.

Norm Referenced Assessment (NRT)

Type of test which tells us where a student stands compared to other students. It helps to determine a student‘s place or rank among a group of similar students. Such kind of test is called norm-referenced test.

Dimensions

- It provides estimate of ability in a variety of skills in much shorter time.
- NRT tends to be general. It measures variety of skills at same time but fails to measure them thoroughly.
- It is hard to make decisions regarding the mastery of student‘s skill in subject.
- It provides estimate of ability in a variety of skills in much shorter time. NRT is much difficult for students to solve. On average only 50% students are able to get an item right in a test.

Topic- 021: Criterion Referenced Assessment (CRT)

This type of test tells us about student’s level of proficiency in or mastery of some skill or set of skills. This is achieved by comparing a student‘s performance to a standard mastery called a criterion. Test that yields such information is called criterion referenced test.

Dimensions

- CRT tends to be specific. It measures particular a set of skill at one time and focus on level of achievement of that skill. CRT gives clear picture regarding the mastery of student‘s skill in subject.
- It measures skill more thoroughly, so naturally it takes more time comparing to NRT in measuring the mastery of said skill.
- Items included in CRT are relatively easier. Around 80% of the students are expected to respond item correctly in the test.
- CRT compares students ‘performance to the standards indicative of mastery.
- Breadth of content sampled is narrow and covers very few objectives.

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Lesson 07

PURPOSE OF TESTING –II

Topic- 022: Characteristics of Criterion Referenced Assessment

Sampled content in CRT is much more comprehensive, usually three or more items are used to cover single objective.

- The meaning of the score does not depend upon on comparison with other scores.
- It flows directly from the connection between the items and the criterion.
- Items are chosen to reflect the criterion behavior. Emphasis is placed upon the domain of relevant responses.
- Number succeeding or failing or range of acceptable performance used.

Example: 90% proficiency achieved, or 80% class reached 90% proficiency.

Topic- 023: Difference between NRT and CRT

Basis of Comparison

- Comparison targets
- Selection of items
- Meaning of success
- Average item difficulty
- Score distribution
- Reported scores

Comparison Targets

In CRT, the examinee‘s performance is compared to an external standard of competence. While in NRT, examinee‘s performance is typically compared to that of other examinees.

Selection of Items

Items included in CRT are of specific nature and designed for the student skilled in particular subjects. In NRT items are of general knowledge nature. Student should be able to answer it, but superficial knowledge is sufficient to respond the item correctly.

Meaning of Success

In CRT, an examinee is classified as a master or non-master. There is no limit to the number of pass or fail. In NRT, examinee‘s opportunity for success is relative to the performance of the other individuals who take the test.

Average Item Difficulty

In CRT, the average item difficulty is fairly high. Examinees are expected to show mastery. In NRT, the average item difficulty is lower. Tests are able to spread out the examinees ‘and provide a reliable ranking.

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Score Distributions

In CRT, a plot of the resulting score distribution will show most of the scores clustering near the high end of the score scale. In NRT, broader spread of scores is expected, with a few examinees earning very low or high scores and many earning medium scores. **Reported Scores**

In CRT, classification of the examinee is measured as master/non-master or pass/fail. In NRT, percentile ranks or scale scores are frequently used.

Topic- 024: Formative Assessment

Formative assessment provides feedback and information during the instructional process, while learning is taking place and occurring. Formative assessment measures student progress, but it can also assess your own progress as an instructor.

Types of Formative Assessment

- Observations during in-class activities; of student's non-verbal feedback during lecture.
- Homework exercises as review for exams and class discussions
- Reflections journals that are reviewed periodically during the semester
- Question and answer sessions, both formal (planned) and informal (spontaneous)
- Conferences between the instructor and student at various points in the semester
- In-class activities where students informally present their results
- Student feedback collected by periodically answering specific question about the instruction and their self-evaluation of performance and progress

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Lesson 08

PURPOSE OF TESTING-III

Topic- 025: Functions of Formative Assessment

Functions of Formative Assessment

- Focus of measurement in formative assessment is predefined segment of instruction.
- Limited sample of learning tasks are addressed.
- The difficulty of item varies with each segment of instruction.
- Formative assessment is conducted periodically during the instructional process.
- Results of formative assessment are used to improve and direct learning through ongoing feedback.

Topic- 026: Summative Assessment

Summative assessment takes place after the learning has been completed and provides information and feedback that sums up the teaching and learning process. Typically, no more formal learning is taking place at this stage, other than incidental learning which might take place through the completion of projects and assignments. Summative assessment is more product-oriented and assesses the final product, whereas formative assessment focuses on the process toward completing the product. Once the project is completed, no further revisions can be made. If, students are allowed to make revisions, the assessment becomes formative.

Types of Summative Assessment

• Examinations (major, high-stakes exams)
• Final examinations (a truly summative assessment)
• Term papers (drafts submitted during the semester would be a formative assessment)
• Projects (project phases submitted at various completion points could be formatively assessed) • Portfolios (could also be assessed during its development as a formative assessment)
• Performances

Topic- 027: Functions of Summative Assessment

- Focus of measurement in summative assessment is on course or unit objectives.
- Broad sample of all objectives is used in summative assessment.
- This type of assessment uses wide range of difficulty while selecting items for the test.
- Summative assessment is done at the end of the unit or the course.
- Most important functions of summative assessment is to assign grade, certification of accomplishment and evaluation of teaching.

Lesson 09

TABLE OF SPECIFICATION

Topic- 028: Table of Specification

One of the tools used by teachers to develop a blueprint for the test is called —table of specification; in the other word table of specification is a technical name for the blue print of the test. It is the first formal step to develop a test.

Concept of Table of Specification

- It helps a teacher in allotting the questions to different content areas and Bloom‘s learning categories in a systematic manner.
- The blueprint is meant to ensure content validity. Content validity is the most important factor in constructing an achievement test. (will be discussed in later unit)
- A unit test or comprehensive exam is based on several lessons and/or chapters in a book supposedly reflecting a balance between content areas and learning levels (objectives).

Two Ways of Table of Specification

A table of specifications consists of a two-way chart or grid relating instructional objectives to the instructional content.

Table of specification performs two important functions

1. It ensures the balance and proper emphasis across all content areas covered by teacher.
2. It ensures the inclusion of items at each level of the cognitive domain of Bloom's Taxonomy.

Learning objectives	Knowledge	Comprehension	Application	Analysis	Synthesis	Evaluation	Weightage %age
1							
2							
3							
4							
5							

Activate

Topic- 029: Concept of Table of Specification

It helps a teacher in allotting the questions to different content areas and Bloom‘s learning categories in a systematic manner.

Learning objectives	Knowledge	Comprehension	Application	Analysis	Synthesis	Evaluation	Weightage %age
1							
2							
3							
4							
5							

Topic- 030: Elements and Appropriateness in Table of Specification

Carey (1988) listed six major elements that should be attended to in developing a table of specifications for a comprehensive end of unit exam:

i. Balance among the goals selected for the exam (weighing objectives)
ii. Balance among the levels of learning (higher order and lower order)
iii. The test format (objective and subjective)
iv. The total number of items
v. The number of test items for each goal and level of learning
vi. The enabling skills to be selected from each goal framework.

A table of specifications incorporating these six elements will result in a "comprehensive posttest that represents each unit and is balanced by goals and levels of learning"

Checklist for Appropriateness of Table of Specification

- Are the specifications in harmony with the purpose of the test?
- Do the specifications indicate the nature and limits of the achievement domain?
- Do the specifications indicate the types of learning outcomes to be measured?
- Do the specifications indicate the sample of learning outcomes to be measured?
- Is the number of test items indicated for the total test and for each subdivision?
- Are the types of items to be used appropriate for the outcomes to be measured?
- Is the difficulty of the items appropriate for the types of interpretation to be made?
- Is the distribution of items adequate for the types of interpretation to be made?
- If sample items are included, do they illustrate the desired attributes?
- Do the specifications, as a whole, indicate a representative sample of instructionally relevant tasks that fits the used to be made of the results?

Topic- 031: Balance Among Learning Objectives and Their Weightin table of specification

In developing a test blueprint first of all it is necessary to select some learning. Objectives and among this list of learning objectives some objectives are more important in sense that more time of instruction is spent on them while some other are less important in terms of time spent on them in classroom so in developing table of specification balance among these learning objectives is important, for this purpose we need to weigh the learning objectives for calculating their relative weightage in test.

Step 1: Instruction Time

To do the calculations for the instruction time for columns of the table of specifications the teacher must use the following formulas for each objective in the table.

Time in class spent on objective (min) / total time for the instruction being examined (min)

Percentage of instruction time= $\frac{\textit{time spent on objective, content, theme (min)}}{\textit{Total time for the instruction being examined (min)}}$

Percentage of instruction time= $\frac{250}{1000}$

Percentage of instruction time = 25%

Step 2: Examine value

Then the instructor should look at the number of test items/score to be allocated to objective/content/theme 1.

Let us assume total marks of the are100. Then 25 marks should be allocated to questions related to objective/content/theme 1. **Step 3**

Percent of instruction time = Percent of examination value (within ±2 percent, if not, redo test) 25 ±2 = 25 ±2

It can be a bit tricky if the total marks of the test are 50. Then 25% of 50 will be 12.5 marks. Point total of questions for objective / total points * on examination = % of examination value

Topic- 032: Balance among Learning Objectives and their Weight in table of specification: Example

We have learnt to give weightage to the content area in a table of specification. Now we will look at an example to develop table of specification practically. Following is the table of specification comprised of topics to be covered in the test and their weightage that represent percentage of marks for each topic.

Topics/Level	Knowledge	Comprehension	Application	Marks
Pakistan Movement Time: (100/500)*100 = 20%				
Geography of Pakistan Time: (150/500)*100 = 30%				
Climate Change Time: (150/500)*100 = 20%				
Industries Time: (50/500)*100 = 10%				
Economy Time: (50/500)*100 = 10%				
Total (Time: 500/Marks: 50)				

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Let us consider that we have to develop a test of 50 marks according to the above discussed table of specification then distribution of marks for each topic is as under.

Topics/Level	Knowledge	Comprehension	Application	Marks
Pakistan Movement Time: (100/500)*100 = 20%				10 (20%)
Geography of Pakistan Time: (150/500)*100 = 30%				15 (30%)
Climate Change Time: (150/500)*100 = 20%				15 (30%)
Industries Time: (50/500)*100 = 10%				5 (10%)
Economy Time: (50/500)*100 = 10%				5 (10%)
Total (Time: 500/Marks: 50)				50 (100%)

Then we have to consider the importance of each topic for cognitive level of questions according to Bloom‘s Taxonomy.

Topics/Level	Knowledge	Comprehension	Application	Marks
Pakistan Movement Time: (100/500)*100 = 20%	5 (50%)	2 (20%)	3 (30%)	10 (20%)
Geography of Pakistan Time: (150/500)*100 = 30%	2 (10%)	6 (40%)	7 (50%)	15 (30%)

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Climate Change Time:		7 (50%)	8 (50%)	15 (30%)
(150/500)*100 = 20%				
Industries Time: (50/500)*100 = 10%	1 (10%)	1 (20%)	3 (70%)	5 (10%)
Economy Time: (50/500)*100 = 10%	1(20%)	1 (20%)	3 (60%)	5 (10%)
Total (Time: 500/Marks: 50)	9 (18%)	17 (34%)	24 (48%)	50 (100%)

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Lesson 10

SELECTION OF TEST

Topic- 033: Selecting Pre-designed

Published test are designed and conducted in such a manner that each and every characteristic is pre planned and known. There are many published tests available for school use. The two most value to the instructional program are:

i. Achievement tests
ii. Aptitude tests

There are hundreds of tests available for each type. Selecting the most appropriate one is important task. In some cases published tests are used by teachers. But more frequently these are used by provincial or national testing programs.

In classrooms most used published tests are:

i. Achievement tests
ii. Reading test

Published tests commonly used by provincial or national testing programs are:

- Aptitude tests
- Readiness tests
- Placement tests

Topic- 034: Standards for Selecting Appropriate Test –I

Test users should select tests that meet the purpose for which they are to be used and that are appropriate for intended population.

- First define the purpose for testing and the population to be tested and select the test accordingly.
- Investigate the potentially useful sources of information, in addition to the test scores, to validate the information provided by tests.
- Read the materials provided by test developers and avoid using tests for which unclear or incomplete information is provided.
- Become familiar with how and when test was developed and tried out.

Topic- 035: Standards for Selecting Appropriate Test –II

Test users should select tests that meet the purpose for which they are to be used and that are appropriate for intended population.

- Read independent evaluations of a test and of possible alternative measures.
- Examine specimen sets, disclosed tests or sample questions directions, answer sheets, manuals and score reports before selecting the tests.

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- Select and use only those tests for which the skills needed to administer the test and interpret scores correctly are available.

Topic- 036: Fairness in Selecting Appropriate Test

- Evaluate the procedures used by test developers to avoid potentially insensitive content or language.
- Review the performance of test takers of different races, gender, and ethnic groups when sample of sufficient size are available.
- Evaluate the extent to which the performance differences may have been caused by inappropriate characteristics of test.
- Use appropriately modified forms of tests or administration procedures for test takers with handicapping conditions.

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Lesson 11

CHARACTERISTICS OF A GOOD TEST-I

Topic- 037: Characteristics of Good Test: Validity, Reliability and Usability

The most essential characteristics of are:

- Validity
- Reliability
- Usability

Validity

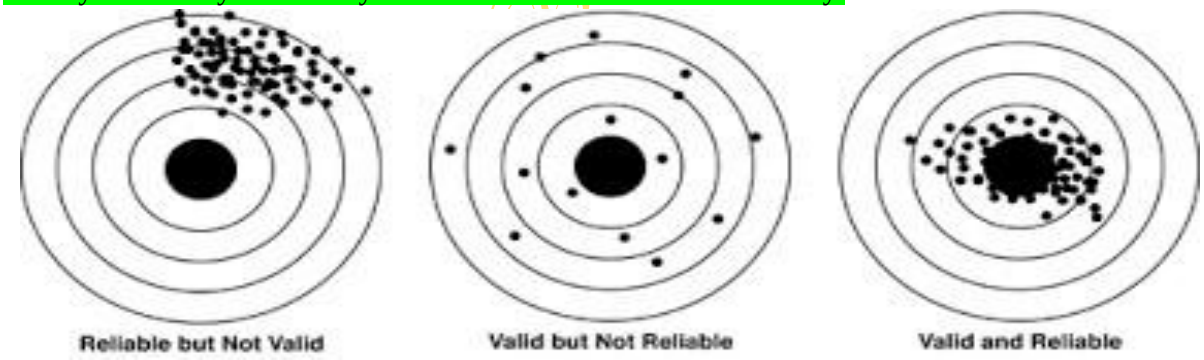
Validity is an evaluation of adequacy and appropriateness of the interpretation and uses of results. It determines if a test is measuring what it intended to measure.

Reliability

Reliability refers to the consistency of assessment results.

Reliability vs. Validity

Reliability of measurement is needed to obtain the valid results, but we can have reliability without validity. Reliability is necessity but not sufficient condition for validity.



Usability

In addition to validity and reliability, an assessment procedure must meet certain practical requirements which include feasibility, administration environment and availability of results for decision makers.

Topic- 038: Nature of Validity

Following are different points that describe the nature of validity.

1. Appropriateness of the interpretation of the results

Validity is referred as ―validity of test but it is in fact validity of the interpretation and use to be made of the results.

2. Validity is a matter of degree

It does not exist an all or none basis. It is best considered in term of categories that specify degree, such as high, moderate or low validity

3. Validity is specific to some particular use or interpretation

No assessment is valid for all purposes. An arithmetic test may have high degree of validity for computational skill and low degree for arithmetical reasoning

4. Validity is a unitary concept

Validity does not have different types. It is viewed as a unitary concept based on different kind of evidences

5. Validity involves an overall evaluative judgment

It requires an evaluation in terms of the consequences of interpretations and uses of assessment results

Topic- 039: Evidences of Validity: Content Validity

Three Evidences of Validity

- Content
- Construct
- Criterion

Meaning

How well the sample of assessment tasks represents the domain of the tasks to be measured.

Procedure

It compares the assessment tasks to the specifications describing the task domain under consideration

Method

1. Classroom instruction determines which intended learning outcomes (objectives) are to be achieved by students.
2. Achievement domain specifies and delimits a set of instructionally relevant learning tasks to be measured by an assessment.
3. Instructional and assessment priorities specifies the relative importance of learning objectives to be assessed.

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Lesson 12

CHARACTERISTICS OF A GOOD TEST-II

Topic- 040: Evidences of Validity: Construct Validity

Meaning

How well a test measures up to its claims? A test designed to measure *depression* must only measure that particular construct, not closely related ideals such as *anxiety* or *stress*.

Procedure

Develop a test framework;

- Defining construct
- Identifying sub-constructs
- Listing indicators of each sub-construct ▪ Writing test items for each indicator

Construct: Essay Writing

Sub-construct	**Meaning/Scope**	**Indicators**
Introduction Paragraph	It introduces the main idea, captures the interest of reader and tells why topic is important.	1. Single sentence called the thesis statement is written 2. Background information about your topic provided 3. Definitions of important terms written
Supporting Paragraphs	Supporting paragraphs make up the main body of your essay	1. List the points about main idea of essay. 2. Write separate paragraph for each supporting point. 3. Develop each supporting point with facts, details, and examples.
Summary Paragraph	Concluding paragraph comes after you have finished developing your ideas.	1. Restate the strongest points of 2. Restate the main idea 3. Give your personal opinion or suggest a plan for action.

Method

There are two methods to confirm construct validity of a test.

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1. Expert judgment

There are experts in the field. For the above example, people who are expert in essay writing will be considered to assess the construct validity of the table and the table will be revised under their guidance.

2. Factor analysis

In this, we group the questions by keeping in view the responses of respondents on them.

Topic- 041: Evidences of Validity: Criterion Validity

Meaning

It demonstrates the degree of accuracy of a test by comparing it with another test, measure or procedure which has been demonstrated is needed to be valid.

Concurrent validity

This approach allows one to show the test is valid by comparing it with an already valid test **Predictive**

It involves testing a group of subjects for a certain construct, and then comparing them with results obtained at some point in the future.

Procedure

It compares assessment results with another measure of performance obtained at a later date (for prediction) or with another measure of performance obtained concurrently (for estimating present status)

Method

The degree of relationship can be described more precisely by statistically correlating the two sets of scores. The resulting *correlation coefficient* provides numerical summary of relationship

Topic- 042: Evidences of Validity: Consequence Validity

Meaning

How well use of assessment results accomplishes intends purposes and avoids unintended effects

Procedure

Evaluate the effects of the use of assessment results on teachers and students. Both, the intended positive effects (e.g., increased learning) and possible unintended negative effects (e.g. dropout of school) need to be evaluated

Considerations

• Does the assessment artificially constrain the focus of student‘s study?

• Does the assessment encourage or discourage exploration and creative modes of expression?

Factors in Test or Assessment Itself

- Unclear directions
- Reading vocabulary and sentence structure too difficult
- Ambiguity
- Inadequate time limits (construct irrelevant variance)
- Overemphasis of easy to access aspects of domain at the expense of important, but hard to access aspects
- Test items inappropriate for the outcomes being measured
- Poorly constructed test items
- Test too short
- Improper arrangement of items
- Identifiable pattern of answers

Lesson 13

CHARACTERISTICS OF A GOOD TEST-III

Topic- 043: Nature of Reliability

Reliability refers to the consistency of measurement.

- Reliability refers to the results obtained with an assessment instrument and not to the instrument itself.
- An estimate of reliability always refers to particular type of consistency (stability, equivalence, internal consistency)
- Reliability is necessary but not sufficient condition for validity.
- Reliability is primarily statistical (range +1 and -1).

Topic- 044: Method of Estimating Reliability 1.

Stability:

Consistency over period of time

2. Equivalence:

Over different forms of assessment

3. Internal consistency:

Within the assessment itself

Determining Reliability by Correlation Methods

In determining reliability, it would be *desirable* to obtain two sets of measures *under identical conditions* and then to compare the results. The reliability coefficient resulting from each method must be interpreted according to type of consistency being investigated.

Method to Estimate Reliability

- Test-Retest (stability)
- Equivalent Forms (equivalence)
- Test- retest with Equivalent Forms (stability and equivalence)
- Split Half (internal consistency)
- Kuder- Richardson(internal consistency)
- Cronbach Alpha (internal consistency)
- Inter-rater reliability (consistency of rating)

Topic- 045: Test-retest Method

- Test-retest method is a measure of stability
- It gives the same test twice to the same group with any time interval between tests, Time interval can range from several minutes to the several years

Test- retest

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September 25	**October 15**
Form A	Form A
1. Item a yes	2. Item a yes
2. Item b no	2. Item b no
3. Item c yes	3. Item c yes

Time interval is key point in this type

- Short interval will provide inflated coefficient of reliability.
- Very long interval will influence results by instability and actual changes in students over time.

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Lesson 14

CHARACTERISTICS OF A GOOD TEST-IV

Topic- 046: Method of Estimating Reliability: Equivalent Form Method

1. Equivalent Forms method
2. Test- Retest with Equivalent Forms

Equivalent Forms method

• The equivalent forms method is a measure of equivalence.

• It gives two forms of the test to the same group in close succession.

September 25	**September 25**
Form A	Form B
1. Item a yes	2. Item d yes
2. Item b yes	2. Item e yes
3. Item c No	3. Item f No

Test- retest with Equivalent Forms

• Test- retest with the equivalent forms is a measure of stability and equivalence.

• It gives two forms of the test to the same group with increased interval between forms

Equivalent Forms Method	
September 25	**September 25**
Form A	Form B
1. Item a	2. Item a
2. Item b	2. Item b
3. Item c	3. Item c
Test- retest with Equivalent Forms	
September 25	**October 15**
Form A	Form B
1. Item a	2. Item a
2. Item b	2. Item b
3. Item c	3. Item c
Score = 82	Score= 74

Topic- 047: Method of Estimating Reliability: Split Half Method

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Split Half Method

• Split half method is a measure of internal consistency.

• It gives test once. Score two equivalent halves of test, correct correlation between halves to fit whole test by spearman –brown formula

Split Half Method		
Sum number of odd items correct	Sum number of even items correct	September 25
Item 1	Item 2	
		1. Item 1
Item 3	Item 4	2. Item 2
Item 5	Item 6	3. Item 3
:	:	4. Item 4
:	:	5. Item 4
:	:	6. Item 4
:	:	
Odd	Even	
Score = 40	Score= 42	Total Score=82

Split half reliabilities tend to be higher than equivalent form reliabilities because split half method is based on the administration of single assessment.

Topic- 048: Method of Estimating Reliability: Kuder-Richardson Method

1. Kuder- Richardson methods and Coefficient Alpha
2. Inter-Rater Method

Kuder- Richardson methods and Coefficient Alpha

• Kuder- Richardson methods and Coefficient Alpha is also a measure of internal consistency.

• It gives test once. Score total test and apply Kuder- Richardson

As with the split half method, these formulas provide an index of internal consistency but do not require splitting the assessment in half for scoring Purposes. One formula KR20 is applicable only when student responses are scored dichotomously (0 or 1). It is the most useful with traditional test items scored correct or incorrect. The generalization of KR20 for assessments that have more than dichotomous, right-wrong scores is called Coefficient Alpha.

Inter-rater Method

• Inter-rater method is a measure of consistency of ratings

• It gives a set of student's responses requiring judgmental scoring to two or more raters and has them independently score the responses

Maha Malik Brain Lab

Lesson 15

TYPES OF ASSESSMENT TOOLS-I

Topic- 049: Anecdotal Records

Many outcomes in the cognitive domain, such as those pertaining to knowledge, understanding, and thinking skills, can be measured by paper pencil tests. But there are still many learning outcomes that require informal observation of natural interactions.

Types of assessment tools

Learning outcomes aspects of development can generally be assessed by:

1. Observing students as they perform and describing or judging that behaviors (Anecdotal record).
2. Asking their peers about them and assessing social relationships (Peer appraisal).
3. Questioning them directly and assessing expressed interests (Self-appraisal).
4. Measuring progress by recorded work (portfolio).

Anecdotal records

Impressions gained through observation are apt to provide an incomplete and biased picture, however, unless we keep an accurate record of our observations. The method to do so is called anecdotal records. Anecdotal records are factual descriptions of meaning incidents and events that the teacher observes.

Topic- 050: Effective Use of Anecdotal Records

One should keep in mind the following points to use anecdotal records effectively.

- Determine in advance what to observe but be alert of unusual behavior.
- Analyze observational records for possible sources of bias.
- Observe and record enough of the situation to make behavior meaningful.
- Make record of the incident as soon after the observation is possible.
- Limit each anecdote to a brief description of a single incident.
- Keep the factual description of the incident and your interpretation of it separate.
- Record both positive and negative behavioral incidents.
- Collect a number of anecdotes on a student before drawing inferences concerning typical behavior.
- Obtain practice in writing anecdotal records.

Topic- 051: Advantages and Limitations of Anecdotal Records

Following are the advantages of anecdotal records.

- It depicts actual behaviors in natural situations.
- It facilitates gathering evidence on the events that are exceptional but significant.
- It is beneficial for the students with less communication skills.

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Limitations of Anecdotal Records

Following are the limitations of anecdotal records.

It takes long time to maintain.

It is subjective in nature.

Anxiety may lead to wrong observation.

Lesson 16

TYPES OF ASSESSMENT TOOLS-II

Topic- 052: Peer Appraisal

In this procedure students rate their peers on the same rating device used by their teacher. It depends on greatly simplified procedures.

Techniques of peer appraisal

There are two widely used techniques in this area are:

1. Guess who
2. Sociometric

Guess who technique

In this technique teacher use a positive or negative behavior of a student as an example. Other students from same group try to guess the statement with that characteristic correctly. Generally behaviors used, for example are positive in nature to avoid any adverse effect on the student pointed out in the example. The guess that technique is based on the nomination method of obtaining peer ratings and is scored by simply counting the number of mentions each student receives on each description.

Sociometric technique

Sociometric technique is a method for assessing the social acceptance of individual students and the social structure group. It is based on students' choice of companion for any group situation. This form was used to measure student's acceptance as seating companions, work companions and play companions.

There are few important principles of sociometric choosing:

1. The choices should be real choices that are the natural part of classroom activities.
2. The basis for the choice and restriction on the choosing should be made clear.
3. All students should be equally free to participate in the activity or situation.
4. The choice of each student make must be kept confidential.
5. The choices should actually be used to organize or rearrange the group.

Topic- 053: Portfolio

Systematic collection of students' work into portfolios can serve a variety of instructional and assessment purposes. The value of portfolios depends heavily on the clarity of purpose the guidelines for the inclusion of materials, and the criteria to be used in evaluating portfolio.

Portfolio of students work

A portfolio is a collection of student work selected to serve a particular purpose, such as documentation of student growth. It is purposeful collection of pieces of student's work.

Key steps in defining and using portfolios

Specify purpose

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Provide guidelines for selecting portfolios

- Define student‘s role in selection and self-evaluation.
- Specify evaluation criteria.

Use portfolios in instruction and communication.

Strengths of portfolios

The can be readily integrated with the instruction.

Provide opportunity to student‘s to show what they can do. Encourage to become reflective learner.

Help in setting goal and self-evaluation

Help teacher and student to collaborate and reflect on student‘s progress.

Effective way to communicate with parents.

Provide mechanism for student centered and student-directed conferences with parents.

Provide concrete examples of student’s development and current skills.

Weaknesses of portfolios

- Can be time consuming to assemble.
- Hard to use in summative assessment.
- Difficult to compare results. ▪ Very low reliability.

Topic- 054: Purpose of Portfolio

Fundamentally, two global purposes for creating portfolios of students work: for student‘s assessment and instruction. It can be used to showcase student‘s accomplishment and document the progress.

Instructional purposes:

When primary purpose is instruction, the portfolio might be used as means of:

1. Helping students develop and refine self-evaluation skills.
2. Providing teacher with more reflecting information regarding students‘progress.
3. Set criteria of excellence between teacher and student.
4. Student directed conferences with parents.
5. Access to student thought process and awareness of standards.
6. Teachers ‘student to communicate with different audience.

Assessment purposes:

When emphasis is on assessment it is important to distinguish between formative and summative roles of assessment.

1. It can be used for formative purposes to measure progress.
2. Basis for certifying accomplishment.
3. For system accountability mechanism.

Current accomplishment and progress

When the focus is on accomplishments, portfolios usually are limited to finished work and may cover only a relatively small period of time. When focus is on demonstrating growth and development the time frame is longer. It will include multiple version of same work over time to measure progress.

Showcase and documentation portfolios

It contains student selected entries. It demonstrates students’ ability to choose his best work which demonstrates his ability to do a task. It intended to provide evidence about breath as well as depth of learning. It needs to be more inclusive and not just limited to special strength of student.

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Finished and working portfolios

It implies that work is complete for specific audience. A job application portfolio is anexample. It is finished product for specific audience. **Guidelines for portfolios entries** Guidelines should specify:

- The uses that will be made of the portfolio.
- Who will have to do it?
- What type of work is appropriate to include.
- What criteria will be used in evaluating the work?
- Should define timeline for the portfolios.
- Minimum and maximum numbers of entries.

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Lesson 17

CREATING FIXED-CHOICE TEST ITEMS: MCQS-I

Topic- 055: Selection of Item in a Test

Selecting Item Format

In accordance with the table of specifications, the teacher decides the item type most appropriate for measuring a stated level of cognitive domains. There are basically two major types of items objective and subjective types. Within objective type items there are: true-false, short answer, multiple choice items, and matching exercises. Which are most appropriate in view the nature of the content, nature of cognitive processes and mental level of the students. The format of the item necessarily proceeds from the test blueprint. The blueprint indicates the kinds of skills and the balance of test content to be measured The format of the item necessarily proceeds from the test blueprint. The blueprint indicates the kinds of skills and the balance of test content to be measured.

The selection of item format should be based on the kinds of skills to be measured and not on some personal like or dislike for a particular item format. The use of multiple-choice questions may make sense for large group testing the knowledge of the mechanics of English. This type of item is not generally appropriate, though, as a direct measure of writing skill. If the intent is to determine whether an examinee can write a clear, coherent essay, then an essay or the free-response format is clearly more appropriate than a multiple-choice format. There is no inherent goodness or badness in any type of question format. The choice must be made on the basis of the behavior to be tested.

Types of Objective Type Items

1. Selection Type
2. Supply Type

In selection type, true-false or alternative form, matching exercises, and multiple choice items are included. Another category of objective type items that is supply type consisted of completion or fill in the blanks and short answer items. The selection of any one or combination of these is made on the basis of their suitability for measuring the desired learning outcome.

Topic- 056: Characteristics of MCQs

The multiple choice items are generally recognized as the most widely applicable and useful types of objective test items that can measure knowledge, comprehension and as well as application level learning outcomes.

Multiple choice items consist of two parts:

i. A Stem
ii. No. of options or alternatives

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Stem

The problem may be stated as a direct question or an incomplete statement and is called the stem of the item.

Alternatives

The list of suggested solutions which may include words, numbers, symbols or phrases are called alternatives (also called choices or options). The correct alternative in each item is called the answer while all incorrect or less appropriate alternatives are called distractors or foiled and the student‘s task is to select the correct or best alternative from all the options.

Format of MCQs

Whether to use a direct question or incomplete statement in the stem depends on several factors. **Direct questions:** the direct questions format is easier to write, is more natural for younger students and is more likely to present a clearly formulated problem.

Example of direct question:

Which of the following cities is the capital of Pakistan? a)
Islamabad
b) Karachi
c) Lahore
d) Quetta

Example of incomplete question:

The capital of Pakistan is: a)
Islamabad
b) Karachi
c) Lahore
d) Quetta

Selection of Format

A common procedure is to start each stem as a direct question and shift to the incomplete statement form only when the clarity of the problem can be retained and greater conciseness achieved.

Format of MCQs

When we get beyond the simple aspects of knowledge, represented by questions of who what when and where variety, answers of varying degree of acceptability are the rule rather than the exception. The question of why variety tends to reveal a number of possible reasons, some of which are better than others. The questions of how variety usually reveals several possible procedures some of which are more desirable then the others. Measures of achievement in these areas then become a matter of selecting the best answer. This type is useful for measuring SLOs that require understanding, application, or interpretation of factual information.

Example of best answer type MCQs

Which of the following factors contributed most to the selection of Islamabad as capital of Pakistan? **a) Central location**
b) Good climate
c) Large population
d) Good highways **Best answer type MCQs**

The best answer type MCQs tend to be more difficult than the correct answer type. This type is used to measure more complex level of learning in students.

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Topic- 057: Uses of MCQs –I

The MCQs is the most versatile types of test item available. It can measure varieties of learning outcomes from simple to complex, and it is adaptable to the most types of matter content.

Despite the wide applicability of the MCQs there are SLOs, such as the ability to organize and present ideas that cannot be measured with any form of selection item. For such skills constructed and restricted response questions are used.

Uses of MCQs in measuring knowledge outcomes

Learning outcomes in the knowledge are so prominent in all school subjects, and MCQs can measure variety of these outcomes. Some of the typical uses are: **Knowledge of Terminology** student can be asked to show their knowledge of particular term by selecting a word that has same meaning as given term.

Example: Which of the following words has same meaning as the word Egress? a) Depress

b) Enter

c) Exit

d) Regress

Example: Which of the following statements best define the word degree? a) An expression of disapproval

b) **An act of leaving an enclosed place**.

c) Proceeding to higher level

Topic- 058: Uses of MCQs –II

Knowledge of Principles

Knowledge of principle is an important learning outcome in most school subjects. MCQs can be constructed to measure knowledge of principle as easily as those designed to measure facts.

Example: The principle of capillary action helps explain how fluids; a) Enter solution of lower concentration

b) Rise in fine tube

c) Escape through small openings

Knowledge of Methods

This includes diverse areas such as knowledge of labs, knowledge of methods underlying communication, computational and performance skills. In some cases we might want to measure knowledge of procedures before we permit students to practice in a particular area. In other cases knowledge of methods may be an important learning outcome in their own right.

Example: To make legislation the prime minister of Pakistan must have the consent of: **a) Parliament**

b) Ministry of law

c) Military command

d) Supreme court

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These uses and their examples have merely scratched the surface of MCQs measuring knowledge outcomes. MCQs can be used to measure the much more complex nature of knowledge. But that is left for more advanced course.

Lesson 18

CREATING FIXED-CHOICE TEST ITEMS: MCQS-II

Topic- 059: Uses of MCQs-III

Another learning outcome basic to all school subjects is the knowledge of specific facts. It provides the basis for developing understanding, thinking skills and other complex learning. MCQs designed to measure specific facts can take many forms, that questions of who, what, when, and where variety are most common.

Example: Who was 1st astronaut to land on moon? a)
Buzz Aldrin
b) Neil Armstrong
c) Yuri Gagarin

Example: What was the name of space shuttle that landed on the moon? **a)**
Apollo
b) Atlas
c) Midas
d) Polaris

Example: When did first man landed on the moon? a)
1962
b) 1966
c) 1967
d) 1969

Topic- 060: Advantages and Limitations of MCQs –I

The MCQ is one of the most widely applicable test items for measuring knowledge, achievement. It can effectively measure various types of knowledge and complex learning outcomes. It is free from some of the common shortcomings which are characteristics of the other test items like ambiguity and vagueness usually associated with short questions.

Example of vague question:
Quid -e -Azam was born in _____.
There can be multiple correct answers for this question. This is poorly constructed item.
Quid -e -Azam was born in: a) Karachi
b) Lahore
c) Peshawar
d) Dehli

Topic- 061: Advantages and Limitations of MCQs –II

MCQs reduces the risk of guessing the correct answer. You have to know the correct answer. There is a high chance of wrong answer if you solely depend on guesses. Example: quid -e -Azam was born in 1867. **True/False**

The student will receive score even if he didn‘t know the correct year of birth Quid –e- Azam if he/she tick n false.

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In which year Quid- e- Azam was born? a)
1867
b) 1876
c) 1878
d) 1887
MCQs items reduced the probability of guessing as compared to other form of item.

Topic- 062: Advantages and Limitations of MCQs –III

Advantages of MCQs

- Ensure objectivity, reliability and validity; preparations of questions with colleagues provide constructive criticism.
- Increase significantly the range and variety of facts that can be sampled in given time.
- Provide precise and unambiguous measurement of the higher intellectual processes.
- Provide detailed feedback for both students and teachers.
- MCQs are easy and rapid to score.

Limitations of MCQs

- Take long time to construct in order to avoid arbitrary and ambiguous questions.
- Require careful preparation to avoid multitude of questions testing only recall.

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Lesson 19

CREATING FIXED-CHOICE TEST ITEMS: MCQS-III

Topic- 063: Suggestions for Constructing MCQs -I

The general applicability and the superior qualities of multiple choice test items are realized most fully when care is taken in their construction. This involves formulating clearly stated problems, identifying plausible alternatives, and removing irrelevant clues to the answer.
The stem of the item should be meaningful by itself and should present a definite problem.
Explanation: often the stem of the test placed in MCQs form are incomplete statements that are make little sense until all the alternatives have been read. This is not MCQs but rather and true false question placed in MCQs form.
Example:
Poor item
South America
a) Is flat, arid country
b) Imports coffee from the United States
c) Has larger population then Europe
d) Was settled by colonist from Spain
Example: **Better item**
Most of the South America was settled by colonists from a)
England
b) France
c) Holland
d) Spain

Topic- 064: Suggestions for Constructing MCQs –II
The item stem should include as much as of the item as possible and should be free of irrelevant material.
Explanation: Clear stem increase the probability of the item as well as reduce the reading time required.
Example: **Poor item**
Most of the Indian subcontinent was settled by colonists from Britain. How would your account for the large number of colonists settling there? a) They are adventurers
b) They were in search of wealth
c) They wanted lower taxes
d) They were seeking religious freedom
Example: **Better item**
Why did Britishers settled in India? a)
For adventures
b) For wealth
c) For lower taxes
d) For religious freedom

Topic- 065: Suggestions for Constructing MCQs –III
Try to avoid the negative statements, unless the significant learning outcome requires it.
Explanation: These avoid the possibility of student overlooking —no, or —least and similar words used in negative items.
Example: **Poor item**
Which of the following cities is not located in north Islamabad? a)
Abbottabad

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b) Gilgit
c) **Lahore**
d) Mingora
Example: **Better item**
Which of the following cities is located in south Islamabad? a)
Abbottabad
b) Gilgit
c) Lahore
d) Mingora

Topic- 066: Suggestions for Constructing MCQs –IV

All alternatives should be grammatically consistent with the stem of item.

Explanation: in the following examples note how the better version results from a change in the alternatives in order to obtain grammatical consistency. The main function of this rule is to prevent irrelevant clues from entering.

Example: **Poor item**
An electric transformer can be used a)
For strong electricity
b) To increase the voltage of alternating current
c) It converts electrical energy into mechanical energy
d) Alternating current is changed to direct current

Example: **Better item**
An electric transformer can be used to a)
Produce strong electricity
b) Increase the voltage of alternating current
c) Convert electrical energy into mechanical energy
d) Change alternating current to direct current

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Lesson 20

CREATING FIXED-CHOICE TEST ITEMS: MCQS-IV

Topic- 067: Suggestions for Constructing MCQs –V

An item should contain only one correct or clearly best answer.

Explanation: Including more than one correct answer in a test item and asking students to select all the correct alternatives has 2 shortcomings.

a. Such items are usually no more than a collection of true and false item presented in MCQ form.
b. The number of alternatives selected as correct answers varies from one student to another.

Example: **Poor item** Pakistan borders on: **a) India**
b) Tajikistan
c) Saudi Arabia

d) China
Example: **Better item**
Pakistan borders on:
a) India T/F
b) Tajikistan T/F
c) Saudi Arabia T/F
d) China T/F

Topic- 068: Suggestions for Constructing MCQs –VI

All distractors should be plausible. The purpose of distractor is to confuse the uniformed.
Explanation: to the student who has not achieved the learning outcome being tested, the distractor should be as attractive as the correct answer. If properly constructed, each distractor will be selected by some students. If the distractor is not selected by anyone, it is not contributing to the functioning of the item and should be eliminated or revised.
Example: **Poor item**
Who wrote national anthem of Pakistan? a)
Allama Iqbal
b) Christopher Columbus
c) **Hafeez Jullandhuri**
d) Ibrar ul Haq
Example: **Better item**
Who wrote national anthem of Pakistan? a)
Allama Iqbal
b) Habib Jalib
c) Hafeez Jullandhuri
d) Munir Niazi

Topic- 069: Suggestions for Constructing MCQs –VII

Verbal association between the stem and the correct answer should be avoided.
Explanation: Frequently a word in the correct answer will provide an alternative clue because it looks or sounds like the word in the stem of the item. However, word similar to those in the stem might be included in the distractors to increase their plausibility. Students who depend on rote memory and verbal association will then led away, from rather than to the correct answer.
Example: **Poor item**
Which of the following agencies should you contact to find about a flood? a)
National flood relief
b) Local radio station
c) Pakistan post
d) Pakistan weather bureau

Example: **Better item**
Which of the following agencies should you contact to find about a flood? a)
Disaster management office
b) Radio station

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c) Post office
d) Weather bureau

Topic- 070: Suggestions for Constructing MCQs –VIII

The relative length of the alternatives should not provide a clue to the answer.

Explanation: the best we can hope for in equalizing the length of the test item's alternatives is to make them approximately equal. But because the correct answer is usually needs to be qualified, it tends to be longer than the distractors unless a special effort is made.

Example: **Poor item**

What is the major purpose of United Nations?

a) To maintain peace among people of the world
b) To establish international law
c) To provide military control
d) To form new governments

Example: **Better item**

What is the major purpose of United Nations?

a) To maintain peace among people of the world
b) To develop new system of international law
c) To provide military control of new nations
d) To establish democratic forms of governments

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